2023 Homework 4
Pre-layout SPICE deliverable

EE3032 Introduction to VLSI Design

13 October 2026

Requested scope

The original HW4 asks for layout of the HW3 three-bit ripple-carry adder, extraction, post-layout delay/power and comparison. The user's explicitly narrower scope stops at the pre-layout `.sp` circuit. Therefore the deliverable is the complete HW3 circuit and independently rerun pre-layout testbench. No layout, DRC, LVS, PEX or post-layout measurement is claimed.

Circuit and evidence

`rca.sp` is byte-equivalent after text line-ending normalization to the verified HW3 netlist: three 28T full adders, 84 transistors total, static CMOS with pMOS bodies at VDD and nMOS bodies at ground. Channel length is 0.18 μm and the size groups are those in the HW3 report. The four external outputs each drive 1 fF. `testbench.sp` uses CIC018 TT, 1.8 V, 25°C, 20 ps input slew, 2 ps maximum timestep, and the 128-vector, 384 ns HW3 workload. `reproduce.py` verifies the netlist hash and launches a separate HSPICE run. All 128 arithmetic states and output voltage bands pass again.

Quantity	Pre-layout result
Verified input states	128/128
Worst single-input delay from HW3	833.526988 ps
Independently rerun VDD power	177.082000 μW
Independently rerun 384 ns energy	67999.487960 fJ

The delay value comes from HW3's exhaustive 896-transition characterization; the independent HW4 rerun verifies the functional workload and power. It does not masquerade as a second exhaustive delay sweep.

Reference comparison

The supplied 2023 HW4 reports and pre-layout files of Lin (110501558), Liao (110501512), Jian (109501570) and Lin (111501559) were inspected. Their layouts and extracted parasitics belong to their different designs. The valid full-adder architecture and rail-body connections are useful references; their post-layout statistics cannot be used as results of this netlist. A physical comparison would require a new layout and extraction, outside the requested scope. The compact independent circuit and exhaustive HW3 proof are retained instead of copying an unrelated PEX file.

Reproduction and files

Run `python reproduce.py -work PATH`; override `-model PATH` and `-hspice PATH` as needed. `vectors.csv` contains all four measured output voltages, expected/observed integer and pass flag. `validation.json` records the separate run and the shared netlist SHA256. The local `D:/cic018.l` model is not redistributed.